

TESTING FOR COMPLETE MARKETS

Part II: Confronting the Theory with Data

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Plan for this lecture

- Complete markets deliver sharp, testable predictions about consumption behavior.
- Today: do those predictions hold up in the data?
 - Classic cross-sectional and between-group tests (1990s)
 - Dynamic predictions: life-cycle inequality and mobility
- Why bother? If markets are not complete:
 - we need models with **incomplete markets**;
 - heterogeneity matters for aggregates (although we still don't know how much).

Are consumers insured against idiosyncratic risk?

- Casual skepticism: there is no formal insurance market for every idiosyncratic shock, and individual consumption clearly moves with individual income.
- But the empirical counterpart of “complete markets” is broader than formal insurance:
 - private insurance and credit markets;
 - public insurance programs;
 - intergenerational and cross-sectional altruism (family, community).
- To determine whether markets are complete, we must test the **quantitative predictions** of the theory against consumption data.

From theory to a testable equation

The planner's problem

- Complete markets $\Rightarrow$ the competitive equilibrium solves a planner's problem for some Pareto weights $\{a^i\}$.
- For general preferences $u(c_t^i, \phi_t^i)$ and a common discount factor β :

$$\max_{\{c^i\}} \sum_i a^i \sum_t \sum_{s^t} \beta^t \pi(s^t) u(c_t^i(s^t), \phi_t^i(s^t)) \quad \text{s.t.} \quad \sum_i c_t^i(s^t) \leq \sum_i y_t^i(s^t)$$

- FOC at each node: $a^i \beta^t \pi(s^t) u_c(c_t^i, \phi_t^i) = \mu_t(s^t)$. Writing $\mu_t(s^t) \equiv \beta^t \pi(s^t) \lambda_t(s^t)$ absorbs discounting and probabilities:

$$a^i u_c(c_t^i, \phi_t^i) = \lambda_t$$

Marginal utility growth is common

- Ratio for the *same* agent i at two dates — a^i cancels:

$$\frac{u_c(c_{t+1}^i, \phi_{t+1}^i)}{u_c(c_t^i, \phi_t^i)} = \frac{\lambda_{t+1}}{\lambda_t}$$

- The right-hand side does **not depend on i** : marginal utility growth is equalized across households.
- This is a clean **testable implication** of complete markets that we can take to the data.

Benchmark: endowment shocks only

- Suppose preferences are CRRA and identical across households: no preference shocks ($\phi_t^i \equiv 1$). Then *measured* consumption growth is

$$\Delta \ln c_{t+1}^i = -\frac{1}{\sigma} \Delta \ln \lambda_{t+1} + \xi_{t+1}^i$$

where ξ_{t+1}^i is measurement error.

- This means that individual consumption growth is **identical across** i up to measurement error, regardless of i 's own endowment realization.

Adding preference shocks

- Keep CRRA, but add a multiplicative preference shock, $u(c, \phi) = \phi \frac{c^{1-\sigma}}{1-\sigma}$, and let ϕ_t^i vary across agents and over time:

$$\Delta \ln c_{t+1}^i = \frac{1}{\sigma} \left[\Delta \ln \phi_{t+1}^i - \Delta \ln \lambda_{t+1} \right] + \xi_{t+1}^i$$

- Consumption growth is **no longer identical** across i — it now moves with $\Delta \ln \phi_{t+1}^i$.
- More generally, every extra source of heterogeneity — discount rates, risk aversion, ... — adds its own term.

A cross-sectional test

Turning the prediction into a test

- Under complete markets, *measured* consumption growth differs across households only because of **preference shifts** and **measurement error**:

$$\Delta \ln c_{t+1}^i = \underbrace{-\frac{1}{\sigma} \Delta \ln \lambda_{t+1}}_{\text{common to all}} + \frac{1}{\sigma} \Delta \ln \phi_{t+1}^i + \xi_{t+1}^i$$

- Idea: pick a household-level variable X_{t+1}^i and ask whether it predicts consumption growth **across households**.
- If X_{t+1}^i is uncorrelated with ξ_{t+1}^i and $\Delta \ln \phi_{t+1}^i$, complete markets imply it is also **uncorrelated with consumption growth**.
- Every extra source of heterogeneity (discount rates, risk aversion, ...) adds its own requirement on X_{t+1}^i .

What makes a good X ?

- A useful X_{t+1}^i must satisfy two conditions:
 - **Relevance**: it would move consumption if insurance failed — otherwise finding no correlation tells us nothing;
 - **Validity**: it is uncorrelated with measurement error and preference shifts — otherwise the test is biased, in either direction.
- The obvious relevant candidate is **income**: an uninsured income shock is exactly what should show up in consumption. But it fails validity:
 - it is largely a *choice* (hours), likely correlated with preference shifts;
 - it includes *transfers*, which respond to households' needs;
 - its measurement error is likely correlated with consumption's.

- Under complete markets, a regression of the form

$$\Delta \ln c_{t+1}^i = \alpha_0 + \alpha_2 X_{t+1}^i + \nu_{t+1}^i$$

should deliver $\alpha_2 = 0$ for any **valid** X_{t+1}^i .

- Regressors (PSID, growth in household *food* consumption, 1980–83): days of work lost to illness, involuntary job loss, weeks of job search after an involuntary job loss, days lost to strikes, involuntary moves.
- The test goes **beyond the CRRA case** above: robust to non-separability and to *any* functional form.

What did Cochrane find?

- Full insurance is **rejected** for long illness (≥ 100 days: 11–14 pp lower consumption growth) and involuntary job loss (24–27 pp lower).
- Not rejected for shorter illness, job search after a job loss, strikes, or involuntary moves — a mixed verdict.
- One can devise stories under which each regressor is correlated with preference shifts. E.g., most households reporting an involuntary job loss report *no* weeks of unemployment — how involuntary was it? Finding a clean X_{t+1}^i is genuinely hard.

A cleaner regressor

From individuals to groups

- Partition the population into groups g defined by characteristics that do not change over time — e.g., birth cohort $\times$ education.
- Average the individual condition (CRRRA, multiplicative ϕ) within each group:

$$\Delta \overline{\ln c}_{g,t+1} = \frac{1}{\sigma} \left[\Delta \overline{\ln \phi}_{g,t+1} - \Delta \ln \lambda_{t+1} \right] + \bar{\xi}_{g,t+1}$$

- With many households per group, idiosyncratic preference shifts and measurement error **average out**: only their group-wide components survive.
- We need an $X_{g,t+1}$ uncorrelated with **group-average** measurement error and preference shifts — and test whether it is also uncorrelated with consumption growth across groups.

- Groups: 5-year birth cohorts $\times$ 4 education levels, US, 1980–90. Consumption from the CEX, wages from the CPS — two surveys, combinable only because the unit of observation is the group.
- Under complete markets, a regression of the form

$$\Delta \overline{\ln c}_{g,t+1} = \alpha_0 + \alpha_2 X_{g,t+1} + e_{g,t+1},$$

with $X_{g,t+1}$ the change in the mean log hourly wage of men in group g , should deliver $\alpha_2 = 0$.

- Their statement of the null: **consumption insurance implies that relative wage movements have no effect on the distribution of marginal utility growth.**
- Still a cross-sectional test, now **across groups.**

Why relative group wages?

- Relative group wages — men's *pretax hourly wage* — keep income's **relevance** and fix its **validity** problems:
- **Measurement**: wages and consumption come from different surveys, so their measurement errors are unrelated.
- **Preferences**: a group's relative wage moves because market-wide forces revalue cohorts and education levels, not because of shifts in the group's own tastes. This is the **identifying assumption**.
- **Choices**: a wage is a price — not hours worked, not transfers — and no household can move its group's average. Group membership is fixed.

What they found: the horizon matters

- Period = a **year**: slope ≈ 0.02 (s.e. 0.15), $R^2 \approx 0$ — no relation.
 - But year-to-year changes are mostly *transitory*: saving and borrowing alone can smooth them, so complete markets and self-insurance *both* predict this.
 - The test has little power here.
- Period = a **decade** (1980–90): slope 0.92 (s.e. 0.12), $R^2 = 0.82$ — relative wage growth passes almost one-for-one into relative consumption growth.
 - Decade-long changes are *persistent*: complete markets predict no response; self-insurance predicts consumption follows them one-for-one.
 - The data side with self-insurance: a “**spectacular failure**” of between-group insurance.

Attanasio and Davis (1996), *Figs. 1–2*.

Dynamic implications

Complete markets freeze the distribution

- Complete markets and self-insurance also make sharply different **dynamic** predictions for the consumption distribution. Back to the level FOC, CRRA, no preference shocks:

$$a^i u_c(c_t^i) = \lambda_t \quad \Rightarrow \quad \ln c_t^i = \frac{1}{\sigma} (\ln a^i - \ln \lambda_t)$$

- Each household's consumption is pinned by its Pareto weight; every shock after date 0 moves everyone proportionally.
- Follow a group with fixed membership — a birth cohort:

$$\text{Var}_i(\ln c_t^i) = \frac{1}{\sigma^2} \text{Var}_i(\ln a^i) \quad \textbf{flat with age}$$

Self-insurance: dispersion accumulates

- Alternative: households only save and borrow at a risk-free rate (Hall, 1978). Consumption is a martingale:

$$\ln c_{t+1}^i \approx \ln c_t^i + \varepsilon_{t+1}^i,$$

with ε_{t+1}^i the news about i 's lifetime resources.

- News is unpredictable, hence uncorrelated with $\ln c_t^i$:

$$\text{Var}_i(\ln c_{t+1}^i) = \text{Var}_i(\ln c_t^i) + \text{Var}_i(\varepsilon_{t+1}^i) \quad \text{rising with age}$$

Deaton and Paxson (1994)

- Follow birth cohorts through repeated cross sections — groups again, not households: US (1980–90), Britain (1969–90), Taiwan (1976–90).
- The within-cohort variance of log consumption **rises with age** in all three: from about 0.2 to about 0.6 over the life cycle — roughly 0.07 (US), 0.08 (Taiwan), 0.10 (Britain) per decade.
- Caveat: with preference shifts, dispersion can move even under complete markets — e.g., if the *spread* of household size across households changes with age. It does: per adult equivalent, the rise in variance per year of age falls sharply in Taiwan ($0.0084 \rightarrow 0.0023$), but barely in the US ($0.0069 \rightarrow 0.0066$).
- Closer to self-insurance than to complete markets.

Deaton and Paxson (1994), [Fig. 4](#) and [Table 1](#).

Complete markets freeze ranks, too

- Back to the level FOC, now with multiplicative preference shocks ($u_c = \phi c^{-\sigma}$):

$$a^i \phi_t^i (c_t^i)^{-\sigma} = \lambda_t \quad \Rightarrow \quad \ln c_t^i = \frac{1}{\sigma} (\ln a^i + \ln \phi_t^i - \ln \lambda_t)$$

- Without preference shocks, a household's rank in the consumption distribution is its rank in Pareto weights: **frozen** — no one ever moves up or down.
- Sharper than the variance: a flat variance can hide households swapping places; frozen ranks cannot.
- With preference shocks, ranks move only if tastes do — e.g., when household composition changes.

- Testing it requires following the *same* households: an Italian panel (1987–95), total **nondurable consumption**.
- Complete markets predict an identity transition matrix across quartiles. In the data (1993–95), about two-thirds of the bottom and top quartiles stay put after two years; only about 40% of the middle quartiles do.
- This survives excluding households whose composition or labor supply changed, and allowing for measurement error and taste shocks, which account for only about a third of the mobility out of the bottom quartile: consumption **mobility** rejects complete markets.

Jappelli and Pistaferri (2006), Table 2.

Conclusions

- Complete markets are rejected **routinely**: in the cross section (Cochrane, Attanasio–Davis) and in the dynamics of the consumption distribution (Deaton–Paxson, Jappelli–Pistaferri).
- Every test leans on assumptions about preferences: unobserved heterogeneity can bias the verdict in either direction (**PS1, Ex. 3**).
- **The road ahead**: models with heterogeneous agents and **incomplete markets**.
- Further reading: **Guvenen (2011)**, a practical survey of heterogeneous-agent macro, including the tests of this lecture.

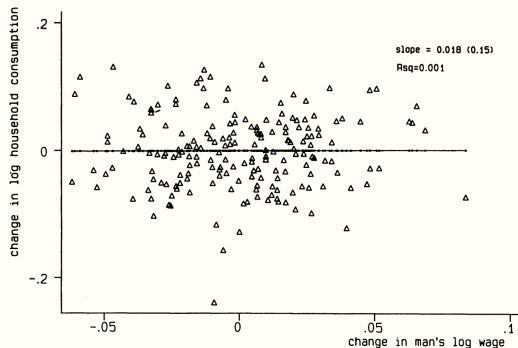
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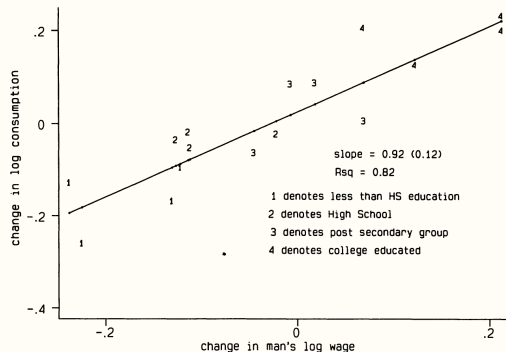
Appendix: figures

Attanasio and Davis (1996): year vs. decade

Annual changes, 1981–90



Ten-year change, 1980–90



Attanasio and Davis (1996), Figs. 1–2. Group-level residuals; groups are 5-year birth cohorts $\times$ 4 education levels.

[◀ Back](#)

Deaton and Paxson (1994): within-cohort variance of log consumption



Deaton and Paxson (1994), Fig. 4. Age effects, with two-standard-error bands.

Jappelli and Pistaferri (2006): consumption transitions, 1993–95

1993–1995				
1993 Quartile	1995 Quartile			
	1st	2nd	3rd	4th
1st	0.67	0.25	0.07	0.01
2nd	0.24	0.43	0.27	0.07
3rd	0.07	0.27	0.43	0.23
4th	0.02	0.05	0.23	0.69

Share of households in each 1993 quartile (row) found in each 1995 quartile (column).
Complete markets: the identity matrix.

Jappelli and Pistaferri (2006), Table 2 (1993–95 panel).

[← Back](#)